for large financial transactions and r3 Corda (https://www.r3.com/corda-platform) for business applications. Recall the layer diagram from chapter 1, shown again in figure 2.1, modified to include smart contract or application logic details. The smart contract is deployed in a sandbox environment and identified by a 160-bit account address like any other participants on the blockchain network. It executes on the virtual machine (VM) on the blockchain node and is identified by an account number, as shown in the figure.

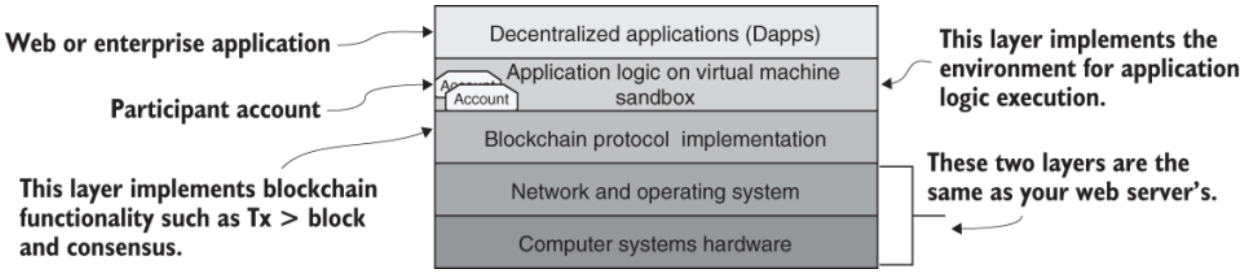


Figure 2.1 Blockchain application stack and layers

2.1.1 Bitcoin transactions versus smart contract transactions

Let's compare a Bitcoin transaction and a smart contract transaction, as shown in figure 2.2, to give you an idea of the difference between currency transactions and noncurrency, application-dependent function calls. As you can see, in Bitcoin, all the transactions are about sending value (`Tx(sendValue)`). In the case of a blockchain that supports smart contracts, a transaction embeds a function implemented by the smart contract. In figure 2.2, this